

The Price of Professionalism

Does Player Market Value Drive Disciplinary Behavior in Football?

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1. Motivation

Football is the world's most popular sport, yet it is also one of its most financially unequal. The past two decades have witnessed an unprecedented commercialization of the game: transfer fees now routinely exceed €100 million, average player salaries at top clubs run into tens of millions per year, and squad valuations at elite sides have surpassed €1 billion. This financial stratification creates a natural experiment that data science is well-placed to examine.

The central question motivating this project is deceptively simple: **does money buy discipline?** More precisely, do teams that invest heavily in their squads (as measured by average player market value, AMV) receive fewer disciplinary sanctions (yellow and red cards) than their lower-budget counterparts? There are two competing intuitions. On one hand, technically superior, expensive players may rely on skill rather than physical aggression, naturally incurring fewer fouls and cards. On the other hand, financial pressure and media scrutiny at elite clubs might push managers to adopt more disciplined tactical systems to protect expensive assets.

Beyond the sporting curiosity, this question has practical implications: clubs, coaches, and analysts who understand the relationship between squad investment and disciplinary behavior can make more informed decisions about squad composition, tactical approach, and youth development philosophy. From a data science perspective, this project offered an opportunity to work with a rich, self-collected cross-league dataset and apply the full pipeline, from data engineering to statistical testing to machine learning, on a real-world problem with non-obvious answers.

2. Hypothesis

The formal hypotheses for this study are:

- H_0 : There is no relationship between a team's average player market value and its disciplinary record.
- H_1 : Teams with higher average player market values receive fewer disciplinary points (negative correlation).

The hypothesis is directional because the financial protection argument provides a clear theoretical basis for expecting the relationship to be *negative* rather than positive. A

team whose average player costs €15M has strong institutional incentives to avoid costly suspensions and injury-inducing tackles that would sideline those investments.

3. Data Source

3.1 Data Collection

Data was collected from two complementary sources to form a unified cross-league panel dataset spanning six major European and one domestic football league across five seasons (2020–21 through 2024–25).

Disciplinary and possession statistics were sourced from FBRef (via StatBomb open data) using the “Share & Export → Get table as CSV” functionality. FBRef provides per-team, per-season breakdowns of yellow cards (**CrđY**), red cards (**CrđR**), and ball possession percentage (**Poss %**).

Squad valuation data was collected manually from Transfermarkt, which provides squad size, total squad market value (in euros), and allows the derivation of average market value per player:

$$\text{AMV} = \frac{\text{Total Squad Value}}{\text{Squad Size}}$$

All values are as reported at the start of each respective season.

3.2 Dataset Characteristics

The final combined dataset contains **583 team-season observations** across 6 leagues and 5 seasons. The leagues covered are: Premier League, La Liga, Serie A, Bundesliga, Ligue 1, and Süper Lig.

Table 1. Dataset summary

Property	Value
Total observations	583
Leagues	6 (PL, La Liga, Serie A, Bundesliga, Ligue 1, Süper Lig)
Seasons	5 (2020–21 through 2024–25)
Primary variables	AMV (€M), CrđY, CrđR, Poss (%)
Engineered features	Discipline Points, CoA Index
Missing values	None (after exclusions)

Two clubs were excluded from the 2022–23 Süper Lig season (Gaziantep FK and Hatayspor) following their mid-season withdrawal in the aftermath of the February 2023 earthquake.

3.3 Engineered Features

Three features were engineered beyond the raw collected variables:

Table 2. Engineered features

Feature	Formula	Description
Discipline Points	$\text{Cr}dY \times 1 + \text{Cr}dR \times 3$	Weighted aggregate disciplinary score per team-season
AMV (€M)	$\frac{\text{Total Value}}{\text{Squad Size}}$	Average per-player market value; normalizes for squad size variation
CoA Index	$DP \times AMV$	Cost of Aggression: monetary cost implied by cards received

The Discipline Points metric weights red cards three times more heavily than yellow cards, reflecting the longer suspension they trigger (typically 3 matches vs. none or 1) and their greater tactical impact.

4. Data Analysis

4.1 Exploratory Data Analysis

The AMV distribution is strongly right-skewed, consistent with the well-documented Pareto-like structure of football player valuations. The median AMV across all 583 observations is €4.4M per player, while the mean is substantially higher due to the long right tail driven by clubs such as Manchester City, Real Madrid, and Paris Saint-Germain. Discipline Points approximate a roughly normal distribution (median = 88) with mild right skew.

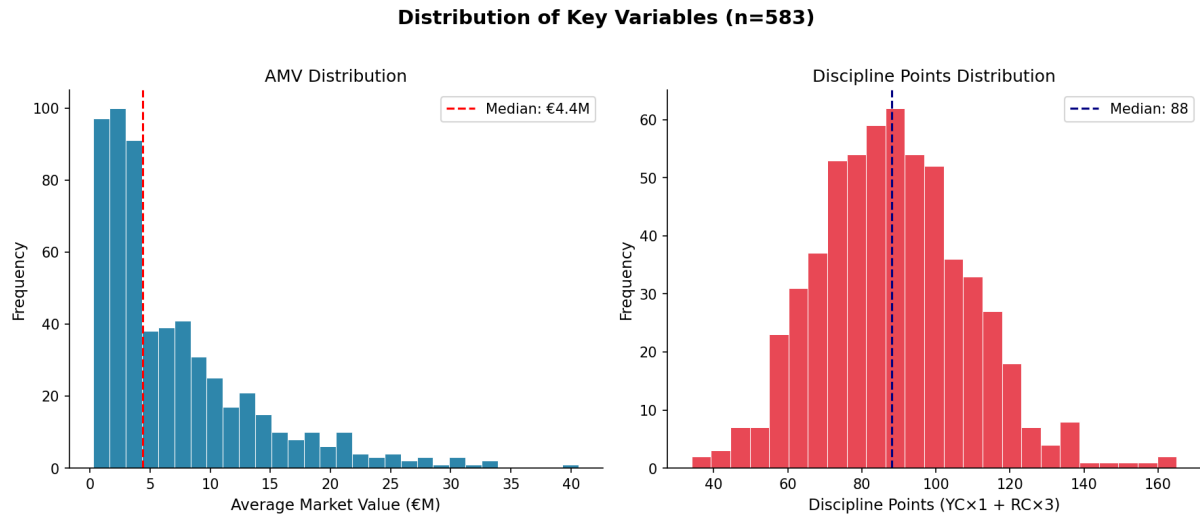


Figure 1. Distribution of AMV and Discipline Points across all 583 team-season observations. AMV is strongly right-skewed; Discipline Points approximate normality.

A correlation heatmap confirms that AMV and Discipline Points are negatively correlated ($r = -0.38$), while possession percentage is positively correlated with AMV ($r = 0.64$) and negatively correlated with Discipline Points ($r = -0.24$), establishing possession as an important mediating variable.

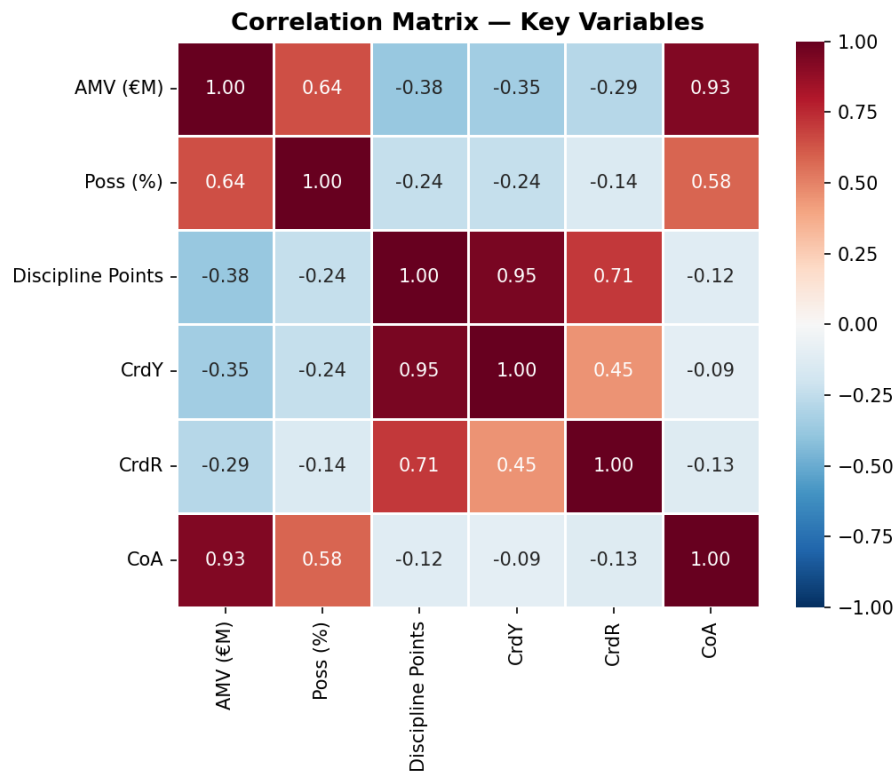


Figure 2. Correlation matrix of key variables. AMV shows the strongest negative relationship with Discipline Points; Possession acts as a partial mediator.

The league-level analysis reveals substantial variation in both AMV and disciplinary behavior across competitions. The Premier League commands the highest median AMV, while La Liga and Süper Lig record the highest mean Discipline Points.

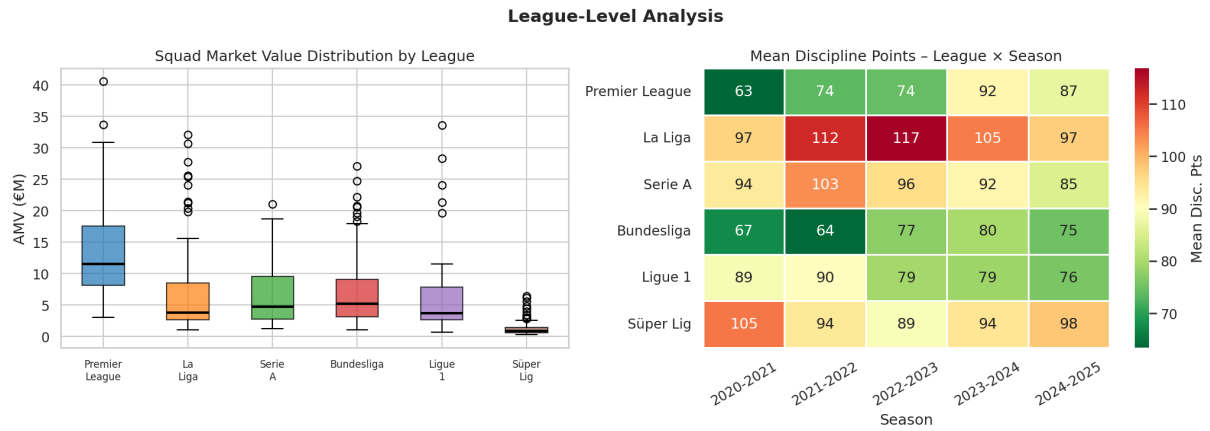


Figure 3. Left: Squad AMV distribution by league. Right: Mean Discipline Points per league per season (2020–25).

A season-by-season trend analysis confirms that the negative AMV–discipline correlation is not a single-season artifact: Pearson r values range from -0.32 to -0.47 across all five seasons, all statistically significant at $p < 0.05$.

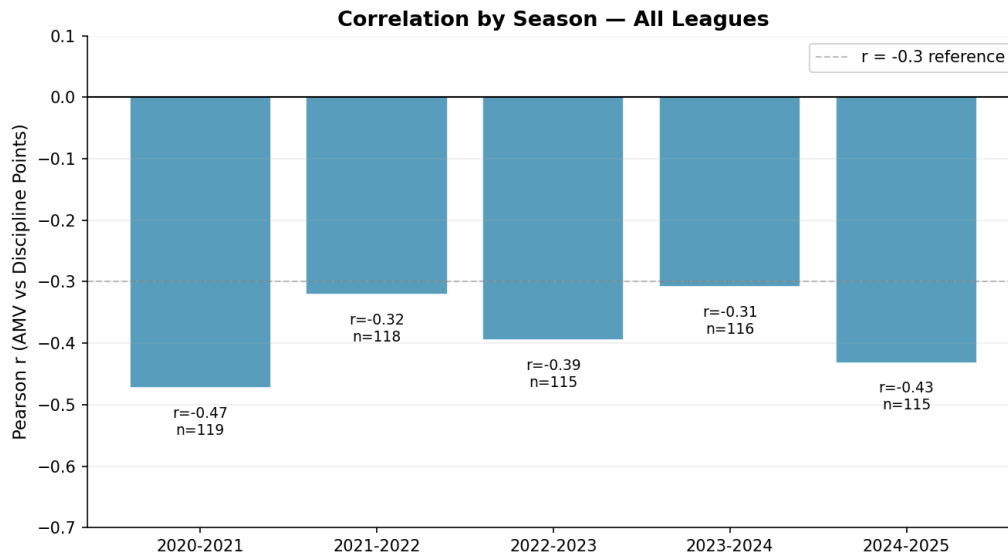


Figure 4. Pearson r between AMV and Discipline Points per season. The negative relationship is consistent across all five seasons.

4.2 Hypothesis Testing

Four formal statistical tests were conducted at a significance threshold of $\alpha = 0.05$.

Test 1: Pearson Correlation.

$$r = -0.377, \quad p < 0.001, \quad 95\% \text{ CI} = [-0.450, -0.301]$$

This constitutes a medium effect size (Cohen 1988). The null hypothesis H_0 is rejected. Figure 5 illustrates the overall negative trend across all 583 observations.

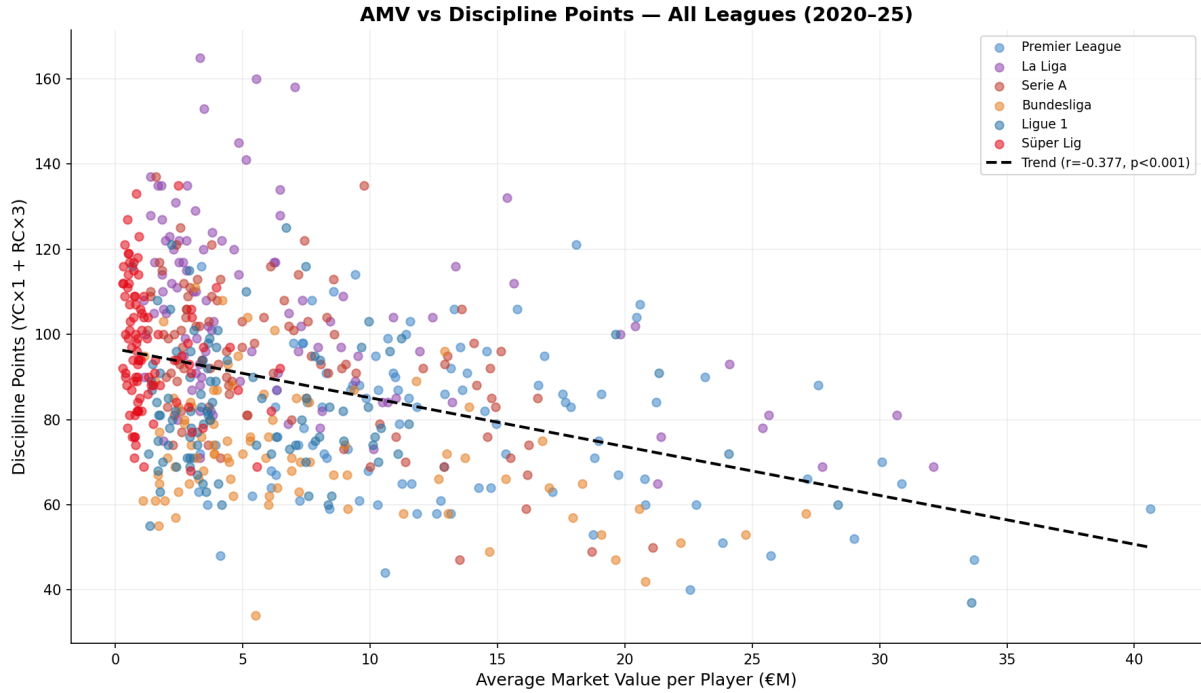


Figure 5. AMV vs Discipline Points across all leagues and seasons (2020–25). The dashed trend line shows $r = -0.377$, $p < 0.001$.

Test 2: Spearman Rank Correlation.

$$\rho = -0.355, \quad p < 0.001$$

The non-parametric result confirms the finding is not driven by distributional assumptions or outliers.

Test 3: Independent Samples t -test. Splitting observations at the median AMV (€4.4M): high-AMV teams average **83** Discipline Points vs. **94** for low-AMV teams, a statistically significant mean difference of 11.1 points ($t = -6.86$, $p < 0.001$, Cohen's $d \approx 0.57$).

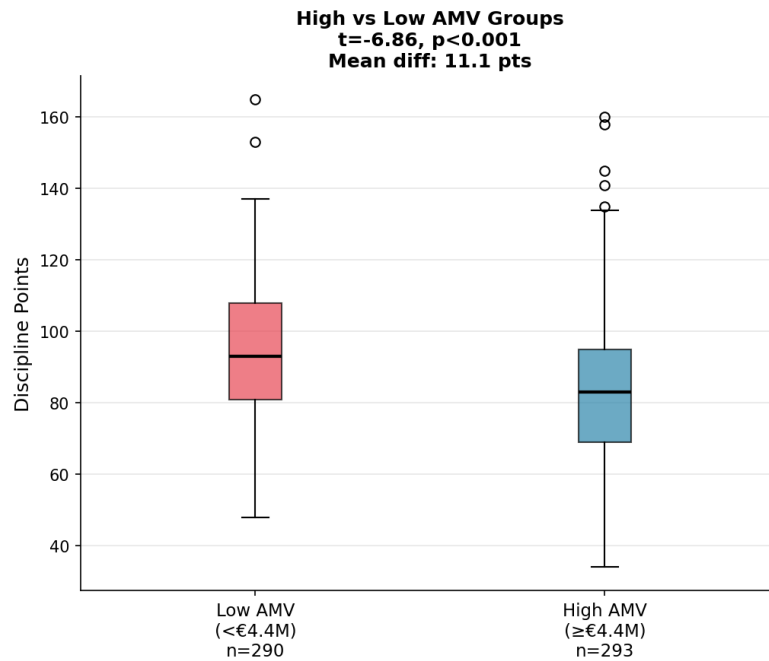


Figure 6. Discipline Points for low-AMV vs high-AMV teams. Mean difference = 11.1 points ($t = -6.86, p < 0.001$).

Test 4: One-way ANOVA. A significant difference in Discipline Points across leagues ($p < 0.001$) confirms that league identity is an important covariate.

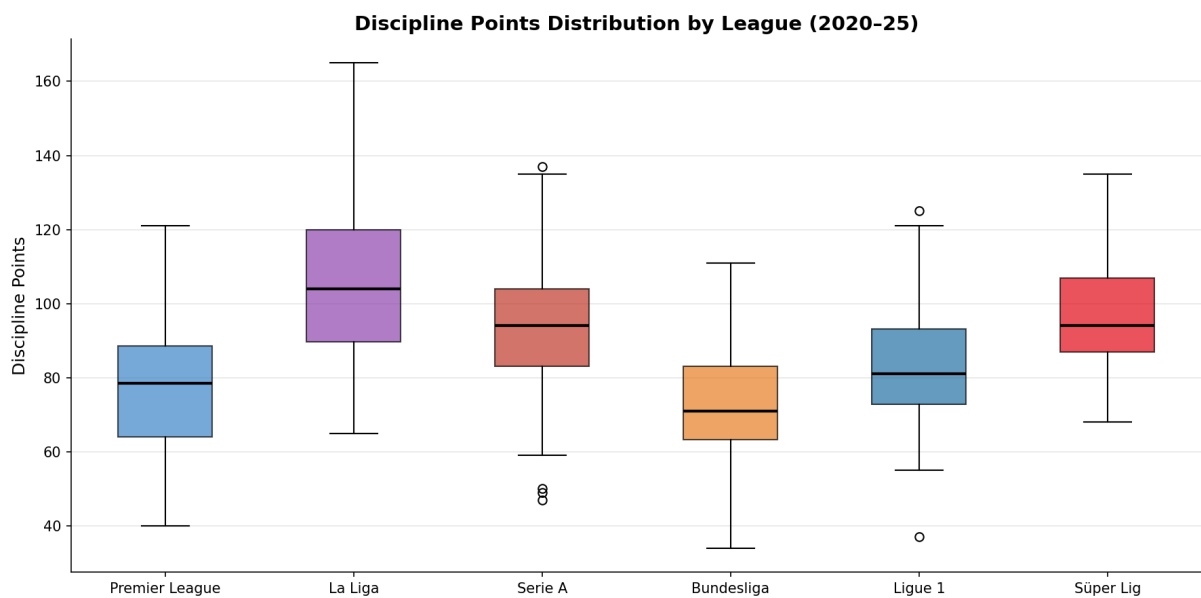
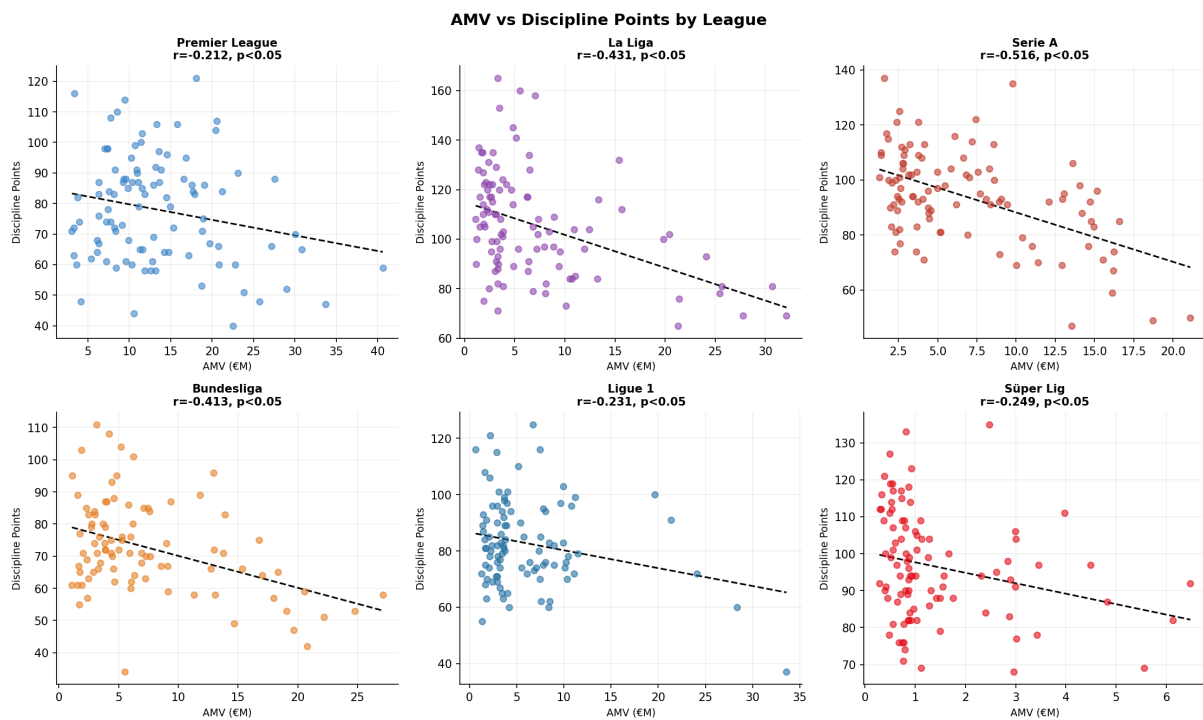


Figure 7. Discipline Points distribution by league. La Liga records the highest median; the Bundesliga and Premier League the lowest, consistent with their higher AMV profiles.

League-by-league breakdown:

Table 3. Pearson and Spearman correlations by league

League	n	Pearson r	Spearman ρ	Significant?
Premier League	100	-0.212	-0.116	Pearson only
La Liga	100	-0.431	-0.414	Yes
Serie A	100	-0.516	-0.417	Yes
Bundesliga	90	-0.413	-0.302	Yes
Ligue 1	100	-0.231	-0.067	Pearson only
Süper Lig	93	-0.249	-0.296	Yes
All Leagues	583	-0.377	-0.355	Yes ($p < 0.001$)

**Figure 8.** AMV vs Discipline Points by league. Serie A shows the strongest correlation ($r = -0.516$); Premier League and Ligue 1 are weaker and likely outlier-driven.

The Premier League and Ligue 1 show significant Pearson but non-significant Spearman correlations, suggesting the linear signal is disproportionately driven by outlier clubs (e.g., Manchester City, PSG) rather than a consistent rank-order trend across the full distribution.

To illustrate this concretely: Manchester City, PSG and other clubs with extreme financial power represent outliers in their respective leagues, combining very high AMV with unusually low Discipline Points. These clubs exert a disproportionate pull on the Pearson correlation, enough to render it statistically significant on their own. Spearman's rank-based structure is resistant to this pull, which explains the divergence between the two

measures in the Premier League and Ligue 1. In leagues where the relationship is more evenly distributed across clubs (Serie A, La Liga, Bundesliga) both measures converge.

Possession is confirmed as a mediating variable ($r = -0.238$ with Discipline Points, $p < 0.001$): wealthier teams control the ball more, structurally reducing their need for defensive fouls.

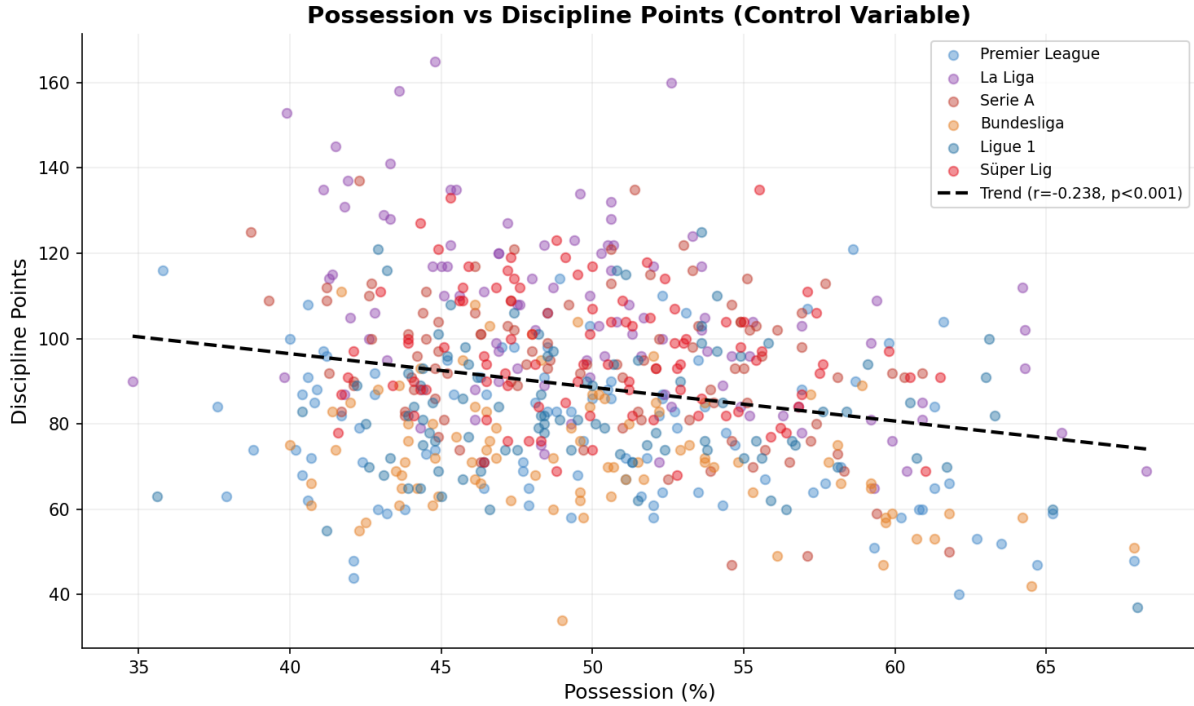


Figure 9. Possession (%) vs Discipline Points. Higher possession is associated with fewer disciplinary sanctions, partially mediating the AMV–discipline relationship.

4.3 Machine Learning Methods

Three ML tasks were conducted. For all tasks, **CrdY**, **CrdR**, and **CoA** were excluded from the feature set to prevent data leakage. Features used: AMV (€M), Possession (%), one-hot encoded League, ordinal-encoded Season. An 80/20 train-test split was used (`random_state=42`).

Regression (predicting Discipline Points). Linear, Ridge, Lasso, and Random Forest regressors were trained. Linear Regression and Ridge ($\alpha = 1.0$) achieved the best generalization:

$$R^2 = 0.4125, \quad \text{MAE} = 13.51, \quad \text{RMSE} = 16.89$$

The models explain $\approx 41\%$ of the variance in discipline points. The remaining 59% is attributable to stochastic in-game factors inherently unpredictable from squad-level features.

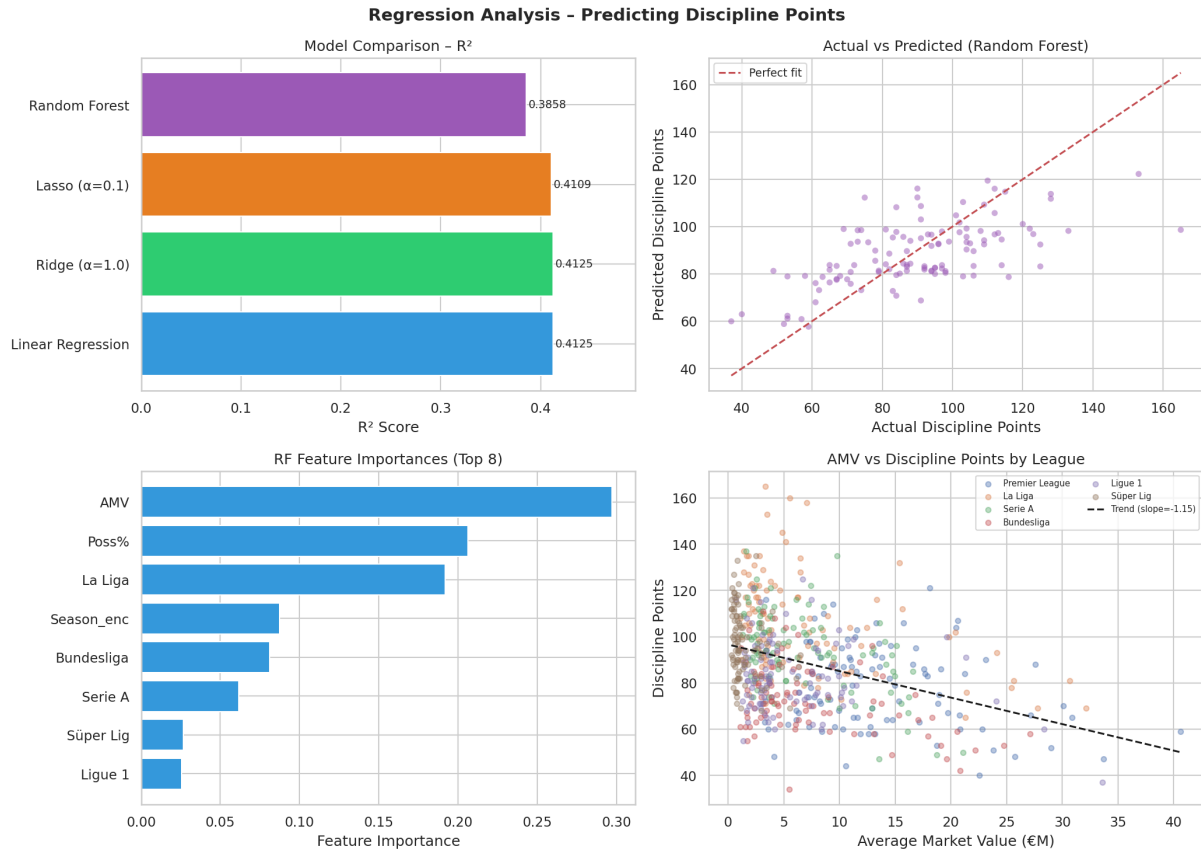


Figure 10. Regression analysis results. Top-left: model comparison by R^2 . Top-right: actual vs predicted (Random Forest). Bottom-left: RF feature importances (AMV and Possession dominate). Bottom-right: regression trend across all leagues (slope = -1.15).

Clustering (team profiling). K -Means with $k = 2$ (silhouette score = 0.426) revealed two dominant team profiles:

Table 4. K -Means cluster profiles ($k = 2$)

Cluster	n	AMV (€M)	Discipline Points
Low Value / Aggressive	430	3.93	94.3
High Value / Disciplined	153	15.68	72.4

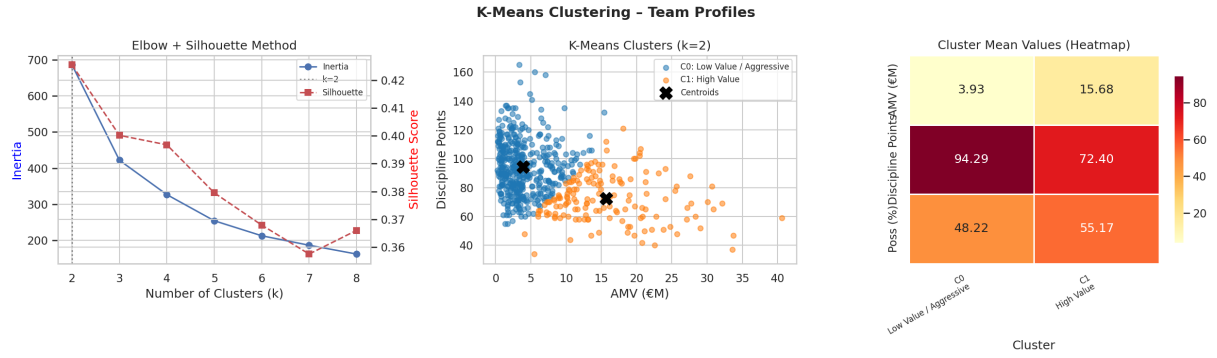


Figure 11. K-Means clustering results. Left: elbow and silhouette analysis selecting $k = 2$. Centre: cluster scatter plot. Right: cluster mean values heatmap showing the 22-point discipline gap.

The 22-point gap in discipline scores, combined with the 7-percentage-point possession differential (48.2% vs. 55.2%), constitutes the project’s most striking finding.

Classification (predicting aggression tier). Discipline Points were binned into three classes: Low (< 79), Medium (79–96), High (> 96). Results:

$$\text{Logistic Regression Accuracy} = 51.3\%, \quad \text{Random Forest CV} = 54.0\% \pm 4.0\%$$

(Random baseline: 33%.) The “High aggression” class was most reliably predicted ($F_1 = 0.65$); “Medium” was hardest to distinguish ($F_1 = 0.29$), as expected for a central category adjacent to both extremes.

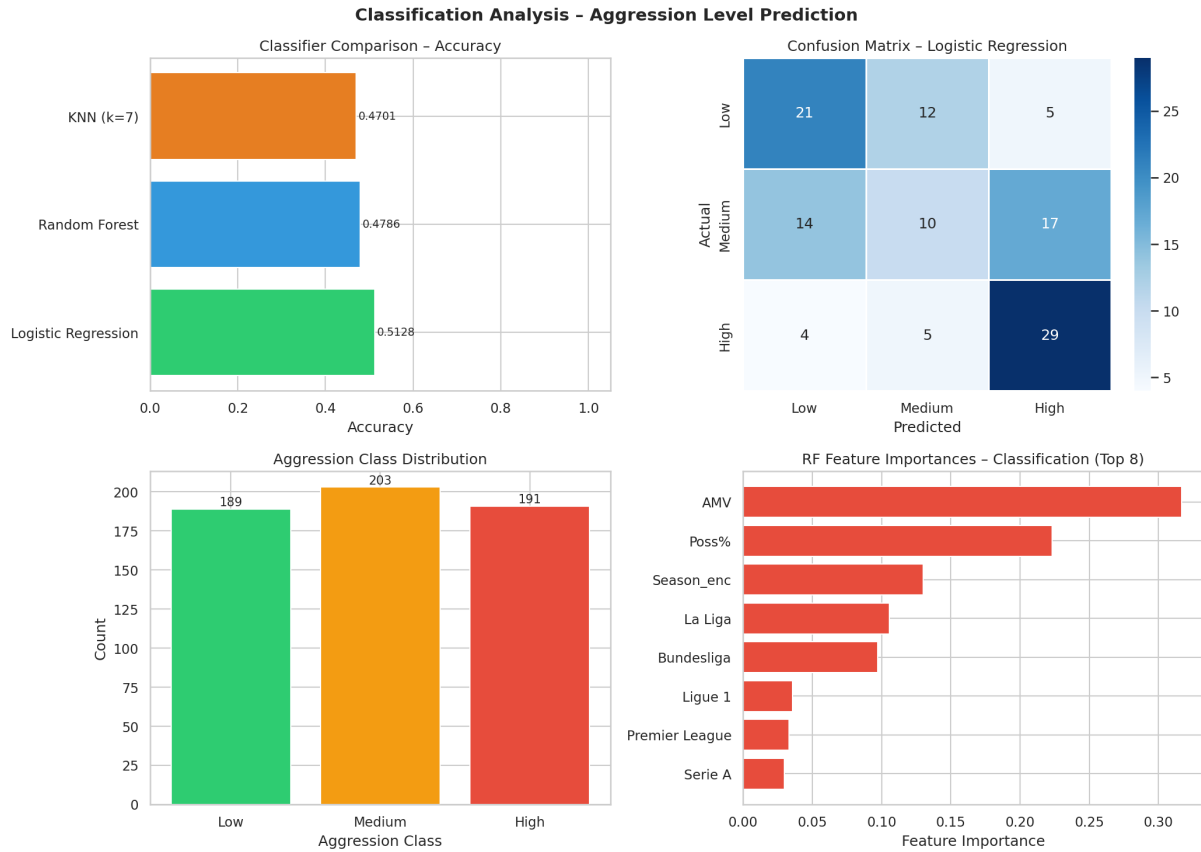


Figure 12. Classification results. Top-left: accuracy comparison across classifiers. Top-right: confusion matrix for Logistic Regression. Bottom-left: class distribution. Bottom-right: RF feature importances (AMV is the dominant predictor).

5. Findings

The evidence consistently and robustly supports H_1 . Key findings:

1. A statistically significant negative correlation exists between average player market value and disciplinary points across all 583 observations ($r = -0.377$, $p < 0.001$; $\rho = -0.355$, $p < 0.001$). The effect size is medium.
2. High-AMV teams receive on average **11 fewer Discipline Points** per season than low-AMV teams ($p < 0.001$, Cohen's $d \approx 0.57$), roughly equivalent to 11 additional yellow cards per season.
3. The relationship holds in 4 of 6 leagues on both Pearson and Spearman measures. In the Premier League and Ligue 1, the linear signal exists but is largely driven by outlier clubs.
4. The relationship is **stable across all five seasons**, with no season showing a positive

or null correlation.

5. *K*-Means clustering identifies two natural team profiles that map almost perfectly onto the hypothesis: a *High Value / Disciplined* cluster and a *Low Value / Aggressive* cluster, separated by 22 Discipline Points and €11.75M in average player value.
6. Possession is a significant mediating variable: wealthier teams control ball possession more effectively, which structurally reduces their need for defensive fouls.
7. Regression models explain $R^2 = 0.4125$ of discipline point variance using squad-level features alone. The residual is driven by in-game randomness beyond the scope of this dataset.

6. Limitations and Future Work

6.1 Limitations

- **Confounding variables:** The dataset does not include managerial philosophy, tactical formation, or individual player profiles (e.g., proportion of defensive midfielders). These factors could account for part of the observed AMV–discipline relationship independently of market value.
- **Cross-league comparability:** Disciplinary standards are not uniform across leagues. Referees in the Süper Lig may apply different thresholds than those in La Liga, introducing systematic league-level noise, confirmed by the significant ANOVA result.
- **Transfermarkt subjectivity:** Market values are community-estimated and may reflect current form, nationality, and media profile in addition to objective footballing ability.
- **Temporal autocorrelation:** The same club appears up to 5 times in the dataset. Observations are not fully independent; a panel data model with fixed effects would be more statistically rigorous.
- **Excluded leagues:** Six leagues limits generalizability. Including lower-division or non-European leagues would test findings across a broader financial and cultural range.

6.2 Future Work

- **Panel data modeling:** A fixed-effects regression controlling for club identity across seasons would eliminate unobserved club-level confounders and yield more causally interpretable estimates.

- **Player-level analysis:** Decomposing squad-level findings to individual player data (cards per 90 minutes vs. market value) would allow a more granular test of the financial protection hypothesis.
- **Causal inference:** An instrumental variable approach or difference-in-differences design around transfer window activity could move the analysis closer to causal claims.
- **Referee fixed effects:** Incorporating match-level referee identity as a control variable would help isolate consistent cross-league disciplinary patterns.
- **Extended time horizon:** Extending the dataset to 10+ seasons would increase statistical power and allow analysis of whether the relationship has changed as transfer fees have inflated.

7. AI Usage Disclosure

In accordance with Sabancı University DSA 210 course policy, the use of generative AI tools in this project is fully disclosed. AI assistance (Claude, Anthropic and Gemini, Google) was used during the brainstorming phase for structuring the research question, during the coding phase for debugging and code review of the Python analysis scripts, and during the writing phase for drafting and proofreading. All analytical decisions, dataset choices, statistical interpretations, and conclusions are the author's own. Full prompt-and-output logs are documented in `AI_Usage_Logs.md` in the project GitHub repository.